

Graduate Level Elementary Particle Physics

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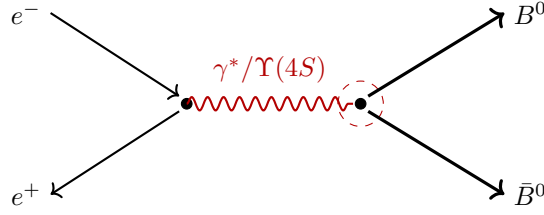
*<https://github.com/yuriphestos/gradphestos-epp>

4 Task 4

4.1 $\Upsilon(4S)$ Resonance and B Meson Oscillations at the Belle-II

Calculate the Breit-Wigner cross section for the $\Upsilon(4S)$ production at e^+e^- collider. Given the projected integrated luminosity of the Belle-II experiment, what is the estimate of the total sample of B mesons to be produced? Given that this is asymmetric collider (look up the energies of e^+ and e^- beams), determine the expected velocities of B mesons. Given the lifetimes of B mesons, what are the minimum requirements for the resolution of the vertex detector, if we are to spatially resolve $B^0 - \bar{B}^0$ oscillations?

We use the Breit-Wigner resonance cross section (PDG 51.1) for $\Upsilon(4S)$ production at an e^+e^- collider:



$$\sigma = \frac{2J+1}{(2S_1+1)(2S_2+1)} \frac{4\pi}{k^2} \left[\frac{\Gamma^2/4}{(E_{cm} - E_0)^2 + \Gamma^2/4} \right] \text{Br}_{\text{in}} \text{Br}_{\text{out}}$$

where $J = 1$ is the spin of the Υ resonance, and the incident e^+e^- polarization states are $(2S_1+1)(2S_2+1) = 2 \times 2 = 4$. For colliding massless leptons, the initial CM momentum is $k = \sqrt{s}/2$. Evaluating exactly at the resonance peak ($\sqrt{s} = E_0 = m_\Upsilon$):

$$\sigma_{\text{peak}} = \frac{3}{4} \frac{4\pi}{(m_\Upsilon/2)^2} \text{Br}(\Upsilon \rightarrow e^+e^-) \text{Br}(\Upsilon \rightarrow \text{Final State}) = \frac{12\pi}{m_\Upsilon^2} \text{Br}_{\text{in}} \text{Br}_{\text{out}}$$

Using $m_{\Upsilon(4S)} = 10.579$ GeV, the cross sections for the total $B\bar{B}$ pair and specifically the neutral $B^0\bar{B}^0$ pair (where $\text{Br}(\Upsilon \rightarrow B^0\bar{B}^0) \approx 48.6\%$) are:

$$\begin{aligned} \sigma_{B\bar{B}} &\simeq \boxed{1.977 \text{ nb}} \\ \sigma_{B^0\bar{B}^0} &\simeq \boxed{0.964 \text{ nb}} \end{aligned}$$

The projected integrated luminosity of the Belle-II experiment is $\mathcal{L} = 50 \text{ ab}^{-1} = 50 \times 10^9 \text{ nb}^{-1}$. The estimated total sample size of B mesons is:

$$\begin{aligned} (\# \text{ of } B\bar{B} \text{ pairs}) &= (50 \times 10^9 \text{ nb}^{-1}) \times (1.977 \text{ nb}) \simeq \boxed{9.88 \times 10^{10}} \\ (\# \text{ of } B^0\bar{B}^0 \text{ pairs}) &= (50 \times 10^9 \text{ nb}^{-1}) \times (0.964 \text{ nb}) \simeq \boxed{4.82 \times 10^{10}} \end{aligned}$$

Asymmetric Collider Kinematics and Vertex Resolution:

Unlike traditional symmetric colliders, the Belle-II collides e^- at 7 GeV (HER) with e^+ at 4 GeV (LER). In the lab frame, the $\Upsilon(4S)$ is produced with a massive net momentum:

$$\begin{aligned} E_\Upsilon &= E_{e^-} + E_{e^+} = 7 + 4 = 11 \text{ GeV} \\ p_\Upsilon &= p_{e^-} - p_{e^+} = 7 - 4 = 3 \text{ GeV} \end{aligned}$$

The invariant mass of the collision is $\sqrt{s} = \sqrt{11^2 - 3^2} = \sqrt{112} \simeq 10.583$ GeV, correctly sitting on the $\Upsilon(4S)$ resonance. The Lorentz boost of the CM frame relative to the lab is:

$$\beta\gamma = \frac{p_\Upsilon}{\sqrt{s}} = \frac{3}{\sqrt{112}} \simeq 0.2835$$

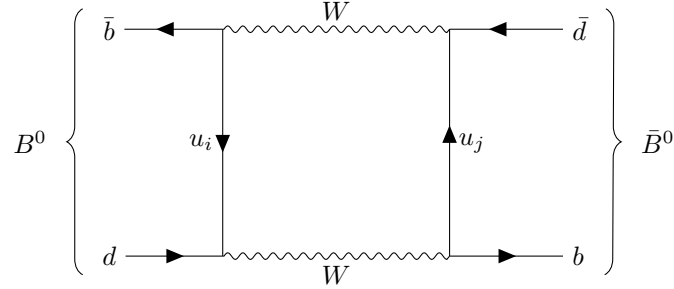
In the $\Upsilon(4S)$ rest frame, the B mesons are produced with very little momentum ($p_B^* \approx 330$ MeV) because the Q -value of the decay is only ~ 20 MeV. Therefore, in the lab frame, the B mesons are overwhelmingly swept forward by the boost of the $\Upsilon(4S)$. We expect the B mesons to travel along the beam axis with the CM velocity $\beta\gamma \simeq 0.2835$.

To observe $B^0 - \bar{B}^0$ oscillations, the detector must resolve the spatial distance Δz between the decay vertices of the two B mesons. Since the characteristic proper time scale of the oscillation is on the order of the B meson lifetime ($\tau_{B^0} = 1.519 \times 10^{-12}$ s), the characteristic spatial separation in the lab frame is:

$$\Delta z = z_1 - z_2 \approx \beta\gamma c\tau_{B^0}$$

Minimum spatial resolution $\langle \ell \rangle \sim (0.2835)(3 \times 10^8 \text{ m/s})(1.519 \times 10^{-12} \text{ s}) \simeq 129 \text{ } \mu\text{m}$

The vertex detector must have a spatial resolution significantly finer than $129 \text{ } \mu\text{m}$ along the beam axis. More precisely, we introduce the mixing frequency $\Delta m_d = 0.5065 \text{ ps}^{-1}$ (PDG), which arises from the box diagram shown below.



The dimensionless mixing parameter is:

$$x_d \equiv \Delta m_d \cdot \tau_{B^0} = (0.5065 \text{ ps}^{-1})(1.519 \text{ ps}) = 0.770$$

Since $x_d < 2\pi$, the B^0 does *not* complete a full oscillation cycle before decaying; instead, it undergoes roughly $x_d/(2\pi) \approx 12\%$ of a full cycle. The “spatial oscillation wavelength” in the lab frame is:

$$\lambda_{\text{osc}} = \frac{2\pi \beta\gamma c}{\Delta m_d} = \frac{2\pi \times 129 \text{ } \mu\text{m}}{0.770} \simeq 1052 \text{ } \mu\text{m}$$

while the “spatial oscillation radian length” is:

$$\ell_{\text{osc}} = \frac{\beta\gamma c}{\Delta m_d} = \frac{\langle \ell \rangle}{x_d} = \frac{129 \text{ } \mu\text{m}}{0.770} \simeq 168 \text{ } \mu\text{m}$$

To resolve the oscillation pattern in the distribution of the vertex separation Δz , the spatial resolution σ_z of the vertex detector must satisfy:

$$\sigma_z \ll \ell_{\text{osc}} \sim 170 \text{ } \mu\text{m}.$$

In practice, one typically requires $\sigma_z \lesssim \ell_{\text{osc}}/5$, giving:

$$\sigma_z \lesssim 30\text{--}50 \mu\text{m}$$

as the minimum requirement on the vertex detector resolution along the beam axis.

The Belle-II VerteX Detector (VXD) consists of two layers of silicon pixel detectors (PXD) and four layers of silicon strip detectors (SVD), achieving an impact parameter resolution of $\sigma_z \sim 15\text{--}50 \mu\text{m}$ depending on track momentum and geometry. This comfortably meets the above requirement.

This analysis explains precisely why B -factories are designed as asymmetric-energy colliders. In a symmetric collider ($E_{e^-} = E_{e^+}$), the CM frame coincides with the lab frame ($\beta\gamma = 0$), and the B mesons would fly only

$$\ell_{\text{sym}} = \beta_B^* \gamma_B^* c \tau_{B^0} \simeq (0.062)(1.002)(3 \times 10^8)(1.519 \times 10^{-12}) \simeq 28 \mu\text{m}$$

which is at the very edge of silicon detector resolution. Moreover, in a symmetric configuration both B mesons fly back-to-back with equal lab speeds, making it impossible to measure the difference Δz in decay positions (the two vertices would be symmetrically displaced by the same amount from the interaction point, providing no handle on the time difference Δt). The asymmetric boost breaks this symmetry and produces a measurable $\Delta z \propto \Delta t$, enabling time-dependent CP-violation measurements:

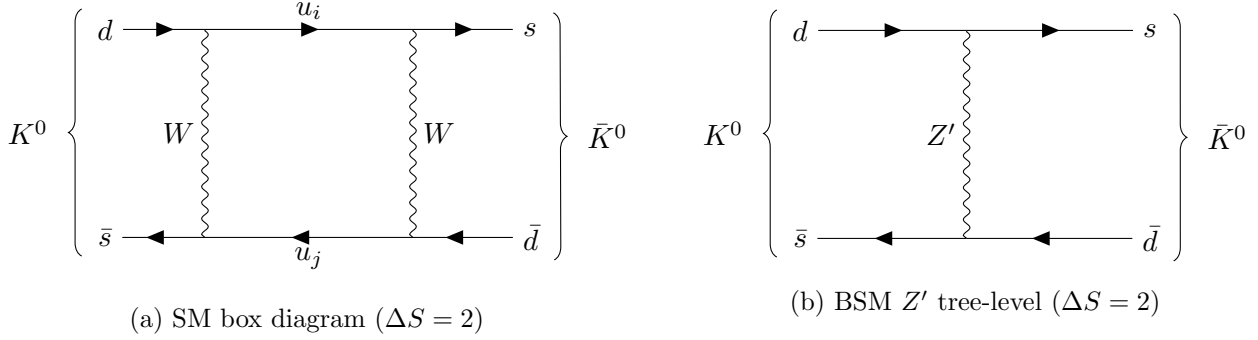
$$\Delta z \approx \beta\gamma c \Delta t$$

4.2 Constraining Superweak New Physics with $K^0 - \bar{K}^0$ Mixing

Using the experimental results for $K_L - K_S$ mass splitting and ϵ_K parameter, derive constraints on the New Physics operator (so-called “superweak” interaction)

$$\mathcal{L}_{NP} = \frac{\text{Re } c + i \text{Im } c}{\Lambda^2} (\bar{s} \gamma^\mu \gamma_5 d) (\bar{s} \gamma_\mu \gamma_5 d) + \text{h.c.}$$

assuming that the current values of Δm_K and ϵ_K are well described by the SM predictions. Compare $1/\Lambda^2$ and G_F if $c \sim O(1)$. Use the so called “vacuum factorization” ansatz to estimate matrix element of the 4-fermion operator inducing the $K^0 - \bar{K}^0$ mixing. This interaction could be induced by a BSM Z' boson, that mediates flavor-changing neutral currents at the tree-level.



Effective Lagrangians for $\Delta S = 2$ Transitions:

The Standard Model generates $\Delta S = 2$ transitions through the box diagram (a) above. After integrating out the W bosons and the internal up-type quarks (dominated by the charm quark at this scale due to the GIM mechanism), the resulting effective Lagrangian takes the $(V-A) \times (V-A)$ form:

$$\mathcal{L}_{SM}^{\Delta S=2} = g [\bar{s} \gamma^\mu (1 - \gamma_5) d] [\bar{s} \gamma_\mu (1 - \gamma_5) d] + \text{h.c.}$$

where the effective coupling is:

$$g = -\frac{G_F^2 m_c^2 \sin^2 \theta_c \cos^2 \theta_c}{16\pi^2}$$

The New Physics operator has a purely axial-vector structure:

$$\mathcal{L}_{NP}^{\Delta S=2} = \frac{\text{Re } c + i \text{Im } c}{\Lambda^2} (\bar{s} \gamma^\mu \gamma_5 d) (\bar{s} \gamma_\mu \gamma_5 d) + \text{h.c.}$$

A key observation is that the SM and NP operators produce the **same** matrix element under vacuum factorization. To see this, expand the SM operator:

$$[\bar{s} \gamma^\mu (1 - \gamma_5) d]^2 = \underbrace{(\bar{s} \gamma^\mu d)^2}_{V \times V} - 2 \underbrace{(\bar{s} \gamma^\mu d)(\bar{s} \gamma_\mu \gamma_5 d)}_{V \times A} + \underbrace{(\bar{s} \gamma^\mu \gamma_5 d)^2}_{A \times A}$$

Under vacuum insertion, only the $A \times A$ term survives: the vector current $\langle 0 | \bar{s} \gamma^\mu d | K^0 \rangle = 0$ vanishes by parity invariance (a pseudoscalar meson cannot couple to a vector current through the vacuum). Therefore, for the purpose of vacuum factorization, $\mathcal{L}_{SM}^{\Delta S=2}$ and $\mathcal{L}_{NP}^{\Delta S=2}$ produce proportional matrix elements—they differ only in their overall coefficients.

Vacuum Factorization of the Hadronic Matrix Element:

We need to evaluate the $\Delta S = 2$ matrix element $\langle K^0 | (\bar{s}\gamma^\mu\gamma_5 d)(\bar{s}\gamma_\mu\gamma_5 d) | \bar{K}^0 \rangle$. The ansatz approximates the four-quark matrix element by inserting a complete set of states and retaining only the vacuum:

$$\langle K^0 | (\bar{s}_\alpha^i \gamma^\mu \gamma_5 d_\beta^j) (\bar{s}_\gamma^k \gamma_\mu \gamma_5 d_\delta^\ell) | \bar{K}^0 \rangle \approx \sum_{\text{color contractions}} \langle K^0 | \bar{s}^i \gamma^\mu \gamma_5 d^j | 0 \rangle \langle 0 | \bar{s}^k \gamma_\mu \gamma_5 d^\ell | \bar{K}^0 \rangle$$

where i, j, k, ℓ are color indices and $\alpha, \beta, \gamma, \delta$ are spinor indices.

Step 1: Kaon decay constant definition.

The axial-vector current matrix element defines the kaon decay constant f_K :

$$\langle 0 | \bar{s}\gamma^\mu\gamma_5 d | K^0(p) \rangle = i f_K p^\mu, \quad \langle K^0(p) | \bar{s}\gamma^\mu\gamma_5 d | 0 \rangle = -i f_K p^\mu.$$

These matrix elements are for color-singlet operators $\bar{s}^i \gamma^\mu \gamma_5 d^i$ (summed over color).

Step 2: Wick contractions and color algebra.

There are two possible ways to contract the quark fields:

- (i) **Direct contraction:** pair $(\bar{s}^i, d^j)$ from the first bilinear with $|K^0\rangle$ and $(\bar{s}^k, d^\ell)$ from the second bilinear with $|\bar{K}^0\rangle$. The color trace gives $\delta^{ij}\delta^{k\ell}$ contracted with the color-singlet meson states, yielding a factor of 1.
- (ii) **Crossed (Fierz) contraction:** pair $(\bar{s}^i, d^\ell)$ with $|\bar{K}^0\rangle$ and $(\bar{s}^k, d^j)$ with $|K^0\rangle$. This requires a Fierz rearrangement of the spinor indices. For the $(A \times A)$ structure $(\gamma^\mu\gamma_5) \otimes (\gamma_\mu\gamma_5)$, the Fierz identity preserves the same Dirac structure. The color trace gives $\delta^{i\ell}\delta^{kj}$, which, when contracted with color-singlet states (each carrying $\delta^{ab}/\sqrt{N_c}$), produces a factor of $1/N_c$.

Combining both contractions:

$$\langle K^0 | (\bar{s}\gamma^\mu\gamma_5 d)(\bar{s}\gamma_\mu\gamma_5 d) | \bar{K}^0 \rangle = \left(1 + \frac{1}{N_c}\right) \times 2 \times \langle K^0 | \bar{s}\gamma^\mu\gamma_5 d | 0 \rangle \langle 0 | \bar{s}\gamma_\mu\gamma_5 d | \bar{K}^0 \rangle$$

where the overall factor of 2 accounts for the two identical bilinears (which operator acts on which meson), and $N_c = 3$ is the number of colors.

Step 3: Evaluating the Lorentz contraction.

Using the decay constant definition with the kaon at rest ($p^\mu = (m_K, \mathbf{0})$):

$$\langle K^0 | \bar{s}\gamma^\mu\gamma_5 d | 0 \rangle \langle 0 | \bar{s}\gamma_\mu\gamma_5 d | \bar{K}^0 \rangle = (-i f_K p^\mu)(i f_K p_\mu) = f_K^2 p^\mu p_\mu = f_K^2 m_K^2$$

Step 4: Final result.

Combining Steps 2 and 3:

$$\boxed{\langle K^0 | (\bar{s}\gamma^\mu\gamma_5 d)(\bar{s}\gamma_\mu\gamma_5 d) | \bar{K}^0 \rangle = 2 \left(1 + \frac{1}{N_c}\right) f_K^2 m_K^2 = \frac{8}{3} f_K^2 m_K^2}$$

The full non-perturbative result is conventionally parameterized as $\langle K^0 | \mathcal{O} | \bar{K}^0 \rangle = \frac{8}{3} f_K^2 m_K^2 \hat{B}_K$, where the “bag parameter” $\hat{B}_K$ encodes deviations from vacuum saturation. Lattice QCD gives $\hat{B}_K \approx 0.72\text{--}0.77$; we set $\hat{B}_K = 1$ in the vacuum factorization approximation throughout.

$K^0 - \bar{K}^0$ Mass Matrix:

In the $\{|K^0\rangle, |\bar{K}^0\rangle\}$ basis, the effective Hamiltonian for neutral kaon mixing takes the 2×2 matrix form:

$$\mathbf{H}_{\text{eff}} = \mathbf{M} - \frac{i}{2} \mathbf{\Gamma} = \begin{pmatrix} m_K - \frac{i}{2} \Gamma_K & M_{12} - \frac{i}{2} \Gamma_{12} \\ M_{12}^* - \frac{i}{2} \Gamma_{12}^* & m_K - \frac{i}{2} \Gamma_K \end{pmatrix}$$

where the diagonal entries are equal by CPT invariance. The off-diagonal dispersive (mass) matrix element M_{12} is generated by the $\Delta S = 2$ Lagrangian:

$$M_{12} \equiv \frac{1}{2m_K} \langle K^0 | \mathcal{H}_{\text{eff}}^{\Delta S=2} | \bar{K}^0 \rangle$$

where the $1/(2m_K)$ is the standard relativistic state normalization factor. The eigenstates $|K_L\rangle$ and $|K_S\rangle$ are obtained by diagonalizing $\mathbf{H}_{\text{eff}}$. In the absence of CP violation, these would coincide with the CP eigenstates:

$$|K_1^0\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle) \quad (\text{CP} = +1), \quad |K_2^0\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle) \quad (\text{CP} = -1)$$

Since CP is violated (the decay $K_L \rightarrow \pi\pi$ has been observed), the states are slightly rotated:

$$|K_S\rangle = \frac{1}{\sqrt{1+|\epsilon_K|^2}}(|K_1^0\rangle + \epsilon_K |K_2^0\rangle)$$

$$|K_L\rangle = \frac{1}{\sqrt{1+|\epsilon_K|^2}}(|K_2^0\rangle + \epsilon_K |K_1^0\rangle)$$

where ϵ_K is a small complex parameter measuring indirect CP violation. The eigenvalues of $\mathbf{H}_{\text{eff}}$ give the physical observables:

$$\Delta m_K \equiv m_{K_L} - m_{K_S} = 2 \text{Re } M_{12} \quad (\text{neglecting } \Gamma_{12} \text{ corrections})$$

$$\epsilon_K \simeq \frac{e^{i\pi/4}}{\sqrt{2}} \frac{\text{Im } M_{12}}{\Delta m_K}$$

where the second relation uses the experimental “superweak phase” $\phi_\epsilon \simeq \arctan(2\Delta m_K/\Delta\Gamma_K) \simeq 43.5^\circ \approx \pi/4$.

SM and NP Contributions to M_{12} :

Using the matrix element from Step 4 and the definition of M_{12} , the full off-diagonal element is $M_{12} = M_{12}^{SM} + M_{12}^{NP}$.

For the SM contribution (using $\mathcal{H} = -\mathcal{L}$):

$$M_{12}^{SM} = \frac{-g}{2m_K} \frac{8}{3} f_K^2 m_K^2 = -\frac{4g}{3} f_K^2 m_K$$

where g is the (negative) SM coefficient, making M_{12}^{SM} positive and real to leading order.

For the NP contribution:

$$M_{12}^{NP} = -\frac{(\text{Re } c + i \text{Im } c)}{2m_K \Lambda^2} \frac{8}{3} f_K^2 m_K^2 = -\frac{4}{3} \frac{(\text{Re } c + i \text{Im } c)}{\Lambda^2} f_K^2 m_K$$

PDG inputs:

G_F	$1.1664 \times 10^{-5} \text{ GeV}^{-2}$
m_c	1.27 GeV
$\sin \theta_c \cos \theta_c$	$\sin \theta_c \approx 0.225, \cos \theta_c \approx 0.974$
f_K	0.1557 GeV
m_K	0.4976 GeV
Δm_K^{exp}	$(3.484 \pm 0.006) \times 10^{-15} \text{ GeV}$
$ \epsilon_K ^{\text{exp}}$	$(2.228 \pm 0.011) \times 10^{-3}$

We can verify the SM prediction for Δm_K . Using $\Delta m_K = 2 |M_{12}^{SM}|$:

$$\Delta m_K^{SM} = 2 |M_{12}^{SM}| = \frac{8}{3} |g| f_K^2 m_K = \frac{8}{3} \frac{G_F^2 m_c^2 \sin^2 \theta_c \cos^2 \theta_c}{16\pi^2} f_K^2 m_K$$

Numerically:

$$|g| = \frac{(1.1664 \times 10^{-5})^2 \times (1.27)^2 \times (0.225)^2 \times (0.974)^2}{16\pi^2} \simeq 6.67 \times 10^{-14} \text{ GeV}^{-2}$$

$$\frac{8}{3} f_K^2 m_K = \frac{8}{3} \times (0.1557)^2 \times 0.4976 \simeq 0.03217 \text{ GeV}^3$$

$$\Delta m_K^{SM} \simeq (6.67 \times 10^{-14})(0.03217) \simeq 2.15 \times 10^{-15} \text{ GeV}$$

This is within a factor of ~ 1.6 of the experimental value $3.484 \times 10^{-15} \text{ GeV}$. The discrepancy is expected since we used only the short-distance charm contribution and set $\hat{B}_K = 1$. Long-distance effects (which are notoriously difficult to compute in the kaon system) account for the remainder.

Constraint from Δm_K (bounds $\text{Re } c$):

Assuming the SM correctly reproduces Δm_K , the NP contribution must not exceed the experimental uncertainty:

$$|\Delta m_K^{NP}| = \frac{8}{3} \frac{|\text{Re } c|}{\Lambda^2} f_K^2 m_K \lesssim \delta(\Delta m_K) = 0.006 \times 10^{-15} \text{ GeV}$$

Setting $|\text{Re } c| \sim 1$:

$$\frac{1}{\Lambda^2} \lesssim \frac{0.006 \times 10^{-15}}{(8/3)(0.1557)^2(0.4976)} = \frac{6.0 \times 10^{-18}}{0.03217} \simeq 1.87 \times 10^{-16} \text{ GeV}^{-2}$$

Therefore:

$$\boxed{\Lambda \gtrsim \sqrt{\frac{1}{1.87 \times 10^{-16}}} \simeq 7.3 \times 10^7 \text{ GeV} \quad (\text{from } \Delta m_K)}$$

Comparing with the Fermi constant:

$$\frac{1/\Lambda^2}{G_F} \simeq \frac{1.87 \times 10^{-16}}{1.166 \times 10^{-5}} \simeq 1.6 \times 10^{-11} \Rightarrow \boxed{\frac{1}{\Lambda^2} \sim 10^{-11} G_F \quad (\text{from } \Delta m_K)}$$

Constraint from ϵ_K (bounds $\text{Im } c$):

The CP-violating parameter ϵ_K probes the imaginary part of M_{12} :

$$|\epsilon_K| \simeq \frac{1}{\sqrt{2}} \frac{|\text{Im } M_{12}|}{\Delta m_K}$$

Assuming the SM perfectly accounts for ϵ_K , the NP contribution must hide within the experimental uncertainty $\delta|\epsilon_K| = 0.011 \times 10^{-3}$:

$$\frac{|\text{Im } M_{12}^{NP}|}{\sqrt{2} \Delta m_K} \lesssim \delta|\epsilon_K|$$

Using $\text{Im } M_{12}^{NP} = (4/3)(\text{Im } c/\Lambda^2) f_K^2 m_K$:

$$\begin{aligned} \frac{4}{3} \frac{|\text{Im } c|}{\Lambda^2} f_K^2 m_K &\lesssim \sqrt{2} \delta|\epsilon_K| \Delta m_K^{\text{exp}} \\ &= \sqrt{2} \times (0.011 \times 10^{-3}) \times (3.484 \times 10^{-15}) \\ &\simeq 5.42 \times 10^{-20} \text{ GeV} \end{aligned}$$

Setting $|\text{Im } c| \sim 1$:

$$\frac{1}{\Lambda^2} \lesssim \frac{5.42 \times 10^{-20}}{(4/3)(0.1557)^2(0.4976)} = \frac{5.42 \times 10^{-20}}{0.01608} \simeq 3.37 \times 10^{-18} \text{ GeV}^{-2}$$

Therefore:

$$\Lambda \gtrsim \sqrt{\frac{1}{3.37 \times 10^{-18}}} \simeq 5.5 \times 10^8 \text{ GeV} \quad (\text{from } \epsilon_K)$$

Comparing with the Fermi constant:

$$\frac{1/\Lambda^2}{G_F} \simeq \frac{3.37 \times 10^{-18}}{1.166 \times 10^{-5}} \simeq 2.9 \times 10^{-13} \Rightarrow \boxed{\frac{1}{\Lambda^2} \sim 10^{-13} G_F \quad (\text{from } \epsilon_K)}$$

The ϵ_K constraint is two orders of magnitude stronger than the Δm_K constraint. This is because ϵ_K is measured to $\sim 0.5\%$ relative precision, and furthermore it probes the imaginary part of M_{12} , which is suppressed in the SM by the Jarlskog invariant—making it exquisitely sensitive to new CP-violating phases.

Summary of Constraints:

Observable	Constrains	$1/\Lambda^2$	Λ
Δm_K	$\text{Re } c$	$\lesssim 10^{-11} G_F$	$\gtrsim 7 \times 10^7 \text{ GeV}$
ϵ_K	$\text{Im } c$	$\lesssim 10^{-13} G_F$	$\gtrsim 5 \times 10^8 \text{ GeV}$

Z' Interpretation and the Flavor Problem:

The superweak operator could be generated at tree-level by a heavy Z' boson with flavor-changing neutral current couplings to the $\bar{s}d$ current (diagram (b) above). If the Z' couples as

$$\mathcal{L}_{Z'} = g'_{sd} Z'_\mu \bar{s} \gamma^\mu \gamma_5 d + \text{h.c.}$$

then integrating out the Z' at tree-level produces:

$$\frac{c}{\Lambda^2} \sim \frac{(g'_{sd})^2}{M_{Z'}^2}$$

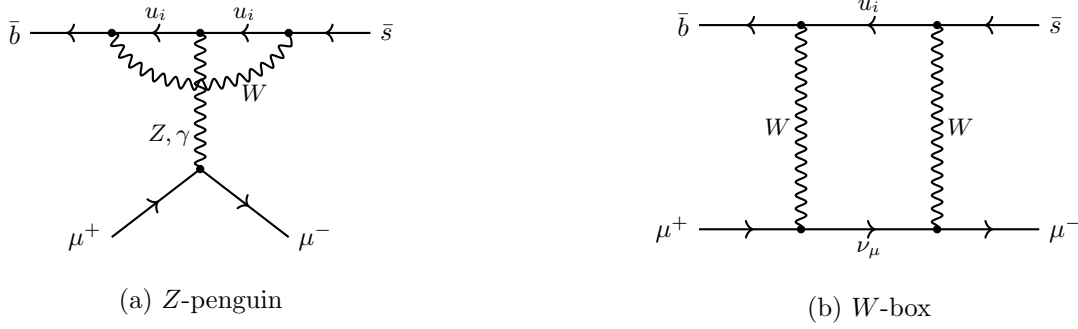
Our constraint from ϵ_K then translates to:

$$\frac{M_{Z'}}{|g'_{sd}|} \gtrsim 5.5 \times 10^8 \text{ GeV}$$

This result has a profound implication for BSM model building. For a Z' at the TeV scale ($M_{Z'} \sim 1 \text{ TeV}$), the flavor-changing coupling must satisfy $|g'_{sd}| \lesssim 2 \times 10^{-6}$ —an extraordinarily small number. Even for $g'_{sd} \sim \alpha/(4\pi) \sim 10^{-3}$ (a loop-level suppression), one would need $M_{Z'} \gtrsim 5 \times 10^5 \text{ GeV}$, far above the electroweak scale. Only if the Z' has a built-in flavor symmetry (e.g., minimal flavor violation, alignment, or discrete symmetries) that suppresses g'_{sd} can a TeV-scale Z' survive the kaon constraints. This is the essence of the **flavor problem** in BSM physics: generic new physics at the TeV scale is excluded by orders of magnitude unless it possesses a highly non-trivial flavor structure.

4.3 The Inami-Lim Loop and $B_s \rightarrow \mu^+ \mu^-$

Take a well-known paper by Inami and Lim that calculates in detail $q_1 \rightarrow q_2 l \bar{l}$ amplitude and reproduce the loop amplitude that depends on m_t, M_W that you can find for example in arXiv:1208.0934. Check the consistency of the SM answer for the $B_s \rightarrow \mu^+ \mu^-$ branching ratio (calculate it using in the same manner we treated $\pi \rightarrow \mu \nu$) and the measured value by the LHCb/CMS/ATLAS collaboration. Treating all constants as known except the top quark mass, what is the indirect limits on m_t that one can extract from this measurement?



Effective Lagrangian and the Inami-Lim Function:

The Z-penguin and W-box diagrams above generate the FCNC process $\bar{b} \rightarrow \bar{s} \mu^+ \mu^-$ at one loop. After evaluating these diagrams and integrating out the heavy degrees of freedom (W , Z , and the internal up-type quarks), the effective Lagrangian for the $B_s \rightarrow \mu^+ \mu^-$ transition takes the form:

$$\mathcal{L}_{eff} = \frac{G_F}{\sqrt{2}} \chi \left[4 \bar{s}_L \gamma_\mu b_L \cdot \tilde{C} \cdot \bar{\mu}_L \gamma^\mu \mu_L \right] + \text{h.c.}$$

where $\chi = \alpha_{em}/(4\pi \sin^2 \theta_W)$ and the factor of 4 is a normalization convention. The Wilson coefficient $\tilde{C}$ encodes the GIM-summed loop amplitude:

$$\tilde{C} = \sum_{j=2}^N V_{js}^* V_{jb} \bar{C}(x_j, x_1 = 0)$$

$$\bar{C}(x_j, x_1 = 0) = \frac{3}{4} \left(\frac{x_j}{x_j - 1} \right)^2 \ln x_j + \frac{1}{4} x_j - \frac{3}{4} \frac{x_j}{x_j - 1}$$

where $x_j = m_{u_j}^2/M_W^2$, and the GIM mechanism ensures that the $x_1 = m_u^2/M_W^2 \rightarrow 0$ contribution cancels. This function is dominated by the t -quark ($x_t = m_t^2/M_W^2 \simeq 4.64$), giving $|\tilde{C}| = |V_{ts}^* V_{tb}| \cdot \bar{C}(x_t, 0)$.

Reproducing $\bar{C}(x, 0) = 2Y_0(x)$:

The standard Inami-Lim function $Y_0(x)$ used in arXiv:1208.0934 is defined as:

$$Y_0(x) = \frac{x}{8} \left[\frac{4-x}{1-x} + \frac{3x}{(1-x)^2} \ln x \right]$$

To show $\bar{C}(x, 0) = 2Y_0(x)$, we manipulate $\bar{C}$ by rewriting $x/(x-1) = -x/(1-x)$:

$$\begin{aligned}
 \bar{C}(x, 0) &= \frac{3}{4} \frac{x^2}{(x-1)^2} \ln x + \frac{x}{4} - \frac{3x}{4(x-1)} \\
 &= \frac{3x^2}{4(1-x)^2} \ln x + \frac{x(x-1) - 3x}{4(x-1)} \\
 &= \frac{3x^2}{4(1-x)^2} \ln x + \frac{x^2 - 4x}{4(x-1)} \\
 &= \frac{3x^2}{4(1-x)^2} \ln x + \frac{x(x-4)}{4(x-1)} \\
 &= \frac{x}{4} \left[\frac{3x}{(1-x)^2} \ln x + \frac{4-x}{1-x} \right] = 2Y_0(x) \quad \checkmark
 \end{aligned}$$

Numerically, with $m_t = 173.2$ GeV and $M_W = 80.38$ GeV ($x_t \simeq 4.64$):

$$Y_0(x_t) \simeq 1.038 \quad \Rightarrow \quad \bar{C}(x_t, 0) = 2Y_0(x_t) \simeq 2.075$$

Decay Amplitude and Helicity Suppression (cf. $\pi \rightarrow \mu\nu$):

Using the vacuum factorization ansatz as in Section 2, the axial-vector current matrix element for the B_s meson is:

$$\langle 0 | \bar{s} \gamma_\mu \gamma_5 b | B_s(p) \rangle = i f_{B_s} p_\mu$$

The $(V-A)$ current $\bar{s}_L \gamma_\mu b_L = \frac{1}{2} \bar{s} \gamma_\mu (1 - \gamma_5) b$ extracts only the axial part when sandwiched with the vacuum:

$$\langle 0 | \bar{s} \gamma_\mu (1 - \gamma_5) b | B_s(p) \rangle = -i f_{B_s} p_\mu$$

Thus, the amplitude becomes:

$$\mathcal{M} = \frac{G_F}{\sqrt{2}} \chi \tilde{C} (-i f_{B_s} p_\mu) [\bar{u}(k_1) \gamma^\mu (1 - \gamma_5) v(k_2)]$$

where k_1 is the μ^- momentum and k_2 is the μ^+ momentum, with $p = k_1 + k_2$.

Helicity suppression (the key step, identical to $\pi \rightarrow \mu\nu$).

The leptonic contraction is:

$$\begin{aligned}
 p_\mu \bar{u}(k_1) \gamma^\mu (1 - \gamma_5) v(k_2) &= \bar{u}(k_1) \not{p} (1 - \gamma_5) v(k_2) \\
 &= \bar{u}(k_1) (\not{k}_1 + \not{k}_2) (1 - \gamma_5) v(k_2)
 \end{aligned}$$

Using the Dirac equation $\bar{u}(k_1) \not{k}_1 = m_\mu \bar{u}(k_1)$ and $\not{k}_2 v(k_2) = -m_\mu v(k_2)$, and noting $(1 - \gamma_5)$ anticommutes past $\not{k}_2$ to give $(1 + \gamma_5)$:

$$\begin{aligned}
 &= m_\mu \bar{u}(k_1) (1 - \gamma_5) v(k_2) + \bar{u}(k_1) (1 + \gamma_5) (-m_\mu) v(k_2) \\
 &= m_\mu \bar{u}(k_1) (1 - \gamma_5) v(k_2) - m_\mu \bar{u}(k_1) (1 + \gamma_5) v(k_2) \\
 &= -2m_\mu \bar{u}(k_1) \gamma_5 v(k_2)
 \end{aligned}$$

This is the **helicity suppression mechanism**: a pseudoscalar meson ($J^P = 0^-$) can only decay into a fermion-antifermion pair if the fermion has non-zero mass, because angular momentum conservation forces the fermion

and antifermion into a spin configuration that is forbidden for massless particles. Hence $\mathcal{M} \propto m_\mu$.

The spin-summed and averaged squared amplitude:

$$\sum_{\text{spins}} |\bar{u} \gamma_5 v|^2 = \text{Tr} [\gamma_5 (\not{k}_1 + m_\mu) \gamma_5 (-\not{k}_2 + m_\mu)] = \text{Tr} [(\not{k}_1 - m_\mu)(\not{k}_2 - m_\mu)]$$

Using $\text{Tr}[\not{k}_1 \not{k}_2] = 4k_1 \cdot k_2$ and $\text{Tr}[\mathbf{1}] = 4$, with $2k_1 \cdot k_2 = m_{B_s}^2 - 2m_\mu^2$:

$$\sum_{\text{spins}} |\bar{u} \gamma_5 v|^2 = 4(k_1 \cdot k_2) + 4m_\mu^2 = 2(m_{B_s}^2 - 2m_\mu^2) + 4m_\mu^2 = 2m_{B_s}^2$$

Collecting all factors (the factor 1/2 from the $(1 - \gamma_5)/2$ projection, the factor of 2 from the Wilson coefficient normalization):

$$\langle |\mathcal{M}|^2 \rangle = 4G_F^2 \chi^2 |\tilde{C}|^2 f_{B_s}^2 m_{B_s}^2 m_\mu^2$$

SM Decay Width:

For the two-body decay of a scalar particle at rest, the partial width is:

$$\Gamma_{\mu\bar{\mu}} = \frac{\langle |\mathcal{M}|^2 \rangle}{8\pi} \frac{|\mathbf{k}_1|}{m_{B_s}^2}$$

The CM momentum of each muon is $|\mathbf{k}_1| = \frac{1}{2}m_{B_s} \sqrt{1 - 4m_\mu^2/m_{B_s}^2}$, so:

$$\begin{aligned} \Gamma_{\mu\bar{\mu}} &= \frac{G_F^2}{2\pi} \chi^2 |\tilde{C}|^2 f_{B_s}^2 m_{B_s}^2 m_\mu^2 \times \frac{1}{2m_{B_s}} \sqrt{1 - \frac{4m_\mu^2}{m_{B_s}^2}} \\ &= \boxed{\frac{G_F^2}{4\pi} \chi^2 |\tilde{C}|^2 f_{B_s}^2 m_{B_s} m_\mu^2 \sqrt{1 - \frac{4m_\mu^2}{m_{B_s}^2}}} \end{aligned}$$

Branching Ratio:

The branching ratio is $\text{Br}(B_s \rightarrow \mu\bar{\mu}) = \tau_{B_s} \times \Gamma_{\mu\bar{\mu}}$. For the t -quark dominated contribution:

$$|\tilde{C}|^2 = |V_{ts}^* V_{tb}|^2 \times (2Y_0)^2 = 4|V_{ts}^* V_{tb}|^2 \times \left\{ \frac{x}{8} \left[\frac{4-x}{1-x} + \frac{3x}{(1-x)^2} \ln x \right] \right\}^2$$

Hence:

$$\text{Br}(B_s \rightarrow \mu\bar{\mu})_{SM} = \boxed{\frac{G_F^2}{\pi} \left(\frac{\alpha_{\text{em}}}{4\pi \sin^2 \theta_W} \right)^2 \tau_{B_s} f_{B_s}^2 m_{B_s} m_\mu^2 \sqrt{1 - \frac{4m_\mu^2}{m_{B_s}^2}} |V_{ts}^* V_{tb}|^2 Y_0^2}$$

PDG and arXiv:1208.0934 inputs:

G_F	$1.1664 \times 10^{-5} \text{ GeV}^{-2}$
$\alpha_{\text{em}}(M_Z)$	$1/127.9$
$\sin^2 \theta_W$	0.2312
f_{B_s}	0.2272 GeV
m_{B_s}	5.3668 GeV
m_μ	0.10566 GeV
τ_{B_s}	$1.466 \times 10^{-12} \text{ s}$
$ V_{ts} $	0.0404
$ V_{tb} $	0.999
m_t	$173.2 \text{ GeV}, M_W = 80.38 \text{ GeV}$

Evaluating at LO:

$$\text{Br}(B_s \rightarrow \mu \bar{\mu})_{SM}^{LO} \simeq \mathbf{3.8 \times 10^{-9}}$$

This leading-order result is $\sim 18\%$ above the full NLO SM prediction of arXiv:1208.0934, which includes QCD corrections (the factor $\eta_Y \approx 0.94$) and the $\Delta\Gamma_s$ correction to the time-integrated branching ratio, yielding:

$$\mathcal{B}_{s,SM}^{(0)} = 3.23 \times 10^{-9} \quad (\text{NLO, arXiv:1208.0934})$$

Comparison with Experiment:

The ATLAS/CMS/LHCb combined measurement (Run 1 + Run 2, 2020) gives:

$$\text{Br}(B_s \rightarrow \mu \bar{\mu})_{\text{exp}} = (2.69_{-0.35}^{+0.37}) \times 10^{-9}$$

This is consistent with the SM prediction within $\sim 1.5\sigma$. (We note that the more recent weighted average, including additional LHCb Run 2 data, gives $(3.34 \pm 0.27) \times 10^{-9}$, in closer agreement with the SM.)

Indirect Constraint on m_t :

Using the parametric formula from Eq. (18) in arXiv:1208.0934:

$$\mathcal{B}_{s,SM}^{(0)} = 3.2348 \times 10^{-9} \times \left(\frac{m_t}{173.2 \text{ GeV}} \right)^{3.07}$$

The power ~ 3.07 reflects the m_t dependence of the Y_0^2 function: since $Y_0 \propto x/8 \sim m_t^2/(8M_W^2)$ for $x \gg 1$, we expect $\text{Br} \propto m_t^4$ at very large m_t , but for physical $x_t \simeq 4.64$ the logarithmic term moderates this to an effective power of ~ 3.07 .

Requiring this to lie within the experimental 1σ band $2.34 \leq \mathcal{B}/10^{-9} \leq 3.06$:

$$2.34 \leq 3.2348 \left(\frac{m_t}{173.2 \text{ GeV}} \right)^{3.07} \leq 3.06$$

Solving:

$$m_t^{(\min)} = 173.2 \times \left(\frac{2.34}{3.2348} \right)^{1/3.07} = 173.2 \times (0.7234)^{0.3257} \simeq 155.9 \text{ GeV}$$

$$m_t^{(\max)} = 173.2 \times \left(\frac{3.06}{3.2348} \right)^{1/3.07} = 173.2 \times (0.9460)^{0.3257} \simeq 170.1 \text{ GeV}$$

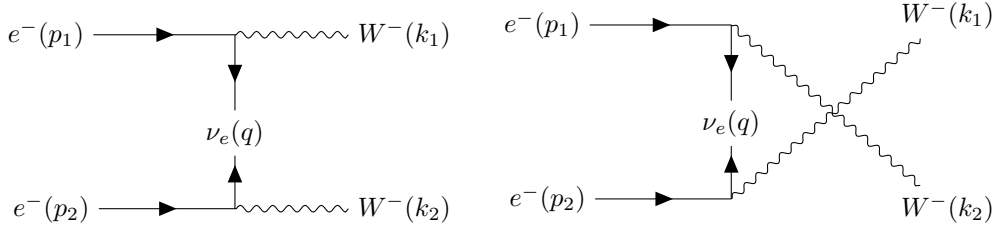
$$\therefore \boxed{155.9 \text{ GeV} \leq m_t \leq 170.1 \text{ GeV}}$$

This range lies below the directly measured value $m_t^{\text{direct}} = 172.69 \pm 0.30 \text{ GeV}$, reflecting the fact that the 2020 combined measurement (2.69×10^{-9}) sits somewhat below the SM prediction, pulling the extracted m_t downward. The tension is at the $\sim 1.5\sigma$ level and illustrates both the power and the current precision limitations of indirect m_t determination from $B_s \rightarrow \mu^+ \mu^-$. Using the more recent experimental average $\text{Br} = (3.34 \pm 0.27) \times 10^{-9}$, one would obtain $m_t \simeq 173 \pm 5 \text{ GeV}$, in excellent agreement with the direct measurement.

4.4 Majorana Neutrinos and Lepton Number Violation: $e^-e^- \rightarrow W^-W^-$

Prove that *if* the Majorana mass of the electron neutrino exists, then $e^-e^- \rightarrow W^-W^-$ process becomes allowed, and calculate the amplitude. (Use B. Kaiser's lecture, if you find it useful.) Given the actual size of neutrino masses, is it a realistic prospect to see $e^-e^- \rightarrow W^-W^-$?

(Mostly following Boris Kayser's Lecture)



A Majorana mass term in the Lagrangian is:

$$\mathcal{L}_m = -\frac{m_{\nu_e}}{2} (\bar{\nu}_e^c \nu_e + \bar{\nu}_e \nu_e^c)$$

This term mixes ν_e and ν_e^c (i.e. particle $\leftrightarrow$ antiparticle), which explicitly violates lepton number by $\Delta L = 2$. The matrix element $\langle \nu_e^c | \mathcal{L}_m | \nu_e \rangle \propto m_\nu \neq 0$, so the transition $\nu_e \leftrightarrow \nu_e^c$ is allowed. This is what enables the $e^-e^- \rightarrow W^-W^-$ process: the upper vertex emits a ν_e (via the standard $e\nu W$ coupling), the Majorana mass insertion converts $\nu_e \rightarrow \nu_e^c$, and the lower vertex absorbs ν_e^c (equivalently, emits a $\bar{\nu}_e$) with the second e^- .

$e\nu_e W$ vertex:

$$-i\frac{g_W}{\sqrt{2}}\bar{\nu}_L\gamma_\mu e_L = i\frac{g_W}{\sqrt{2}}\bar{e}_R\gamma_\mu \nu_R$$

ν_e Propagator:

$$\frac{i(\not{q} + m_\nu)}{q^2 - m_\nu^2} \rightarrow \frac{i(\not{q} + m_\nu)}{q^2}$$

The Dirac structure of the t -channel amplitude reads:

$$\underbrace{\bar{v}(p_1)\gamma_\mu\left(\frac{1+\gamma_5}{2}\right)}_{\text{upper vertex}}(\not{q} + m_\nu)\gamma_\lambda\underbrace{\left(\frac{1-\gamma_5}{2}\right)u(p_2)}_{\text{lower vertex}}$$

where $P_R = (1 + \gamma_5)/2$ comes from the charge-conjugated upper vertex and $P_L = (1 - \gamma_5)/2$ from the standard lower vertex. Note the ordering: P_R acts on the propagator from the left, P_L acts on $u(p_2)$ from the right, and they are separated by γ_λ .

Step 1: The $\not{q}$ term vanishes.

Consider the $\not{q}$ piece:

$$\bar{v}\gamma_\mu(1+\gamma_5)\not{q}\gamma_\lambda(1-\gamma_5)u$$

Using $\gamma^\alpha\gamma_5 = -\gamma_5\gamma^\alpha$, we push $(1 + \gamma_5)$ past $\not{q}$:

$$(1 + \gamma_5)\not{q} = \not{q}(1 - \gamma_5)$$

so the expression becomes:

$$\bar{v} \gamma_\mu \not{q} (1 - \gamma_5) \gamma_\lambda (1 - \gamma_5) u$$

Now, push the first $(1 - \gamma_5)$ past γ_λ :

$$(1 - \gamma_5) \gamma_\lambda = \gamma_\lambda (1 + \gamma_5)$$

giving:

$$\bar{v} \gamma_\mu \not{q} \gamma_\lambda \underbrace{(1 + \gamma_5)(1 - \gamma_5)}_{=0} u = 0 \quad \checkmark$$

$$\boxed{(1 + \gamma_5)(1 - \gamma_5) = 1 - \gamma_5^2 = 0 \implies \not{q} \text{ term vanishes identically}}$$

This holds **regardless of** m_e —it is a pure consequence of the opposite chirality projectors at the two vertices being brought together by the anti-commutation of γ_5 through $\not{q}$ and γ_λ .

Step 2: The m_ν term survives.

For the mass piece, the scalar m_ν commutes freely:

$$m_\nu \bar{v} \gamma_\mu (1 + \gamma_5) \gamma_\lambda (1 - \gamma_5) u$$

Now, push $(1 + \gamma_5)$ past γ_λ :

$$(1 + \gamma_5) \gamma_\lambda = \gamma_\lambda (1 - \gamma_5)$$

giving:

$$m_\nu \bar{v} \gamma_\mu \gamma_\lambda \underbrace{(1 - \gamma_5)(1 - \gamma_5)}_{=2(1 - \gamma_5)} u = 2m_\nu \bar{v} \gamma_\mu \gamma_\lambda (1 - \gamma_5) u \neq 0$$

The key difference from Step 1: for the $\not{q}$ piece, the $\not{q}$ carries a Lorentz index that flips the chirality projector when commuted through (turning P_R into P_L), so both projectors end up as $P_R P_L = 0$. For the m_ν piece, the scalar mass does not flip the projector, so we get $P_L P_L = P_L \neq 0$.

$$\boxed{m_\nu \text{ term} = 2m_\nu \bar{v} \gamma_\mu \gamma_\lambda (1 - \gamma_5) u \neq 0 \implies \mathcal{M} \propto m_\nu}$$

If $m_\nu = 0$, the amplitude vanishes identically. A non-zero Majorana mass is both necessary ($\Delta L = 2$ violation and chirality flip) and sufficient (generates the amplitude) for $e^-e^- \rightarrow W^-W^-$ to occur.

Matrix Element (t, u channels):

Combining the two vertices with the Majorana propagator (only the m_ν piece contributes):

$$\begin{aligned} \mathcal{M}_t &= \left(\frac{-g_W^2}{2} \right) \bar{e}^c(p_1) \varepsilon_1^\mu \gamma_\mu \left(\frac{1 + \gamma_5}{2} \right) \left(\frac{\not{q} + m_\nu}{q^2} \right) \varepsilon_2^\lambda \gamma_\lambda \left(\frac{1 - \gamma_5}{2} \right) e(p_2) \\ &= \left(\frac{-g_W^2 m_\nu}{8t} \right) \varepsilon_1^\mu \varepsilon_2^\lambda [\bar{v}_{e^c} \gamma_\mu (1 + \gamma_5) \gamma_\lambda (1 - \gamma_5) u_e] \end{aligned}$$

where $t = q^2 = (p_1 - k_1)^2$. The u -channel is obtained by $k_1 \leftrightarrow k_2$ (and $\varepsilon_1 \leftrightarrow \varepsilon_2$):

$$\mathcal{M}_u = \left(\frac{-g_W^2 m_\nu}{8u} \right) \varepsilon_2^\mu \varepsilon_1^\lambda [\bar{v}_{e^c} \gamma_\mu (1 + \gamma_5) \gamma_\lambda (1 - \gamma_5) u_e]$$

where $u = (p_1 - k_2)^2$. Using Mandelstam variables ($s + t + u = 2m_W^2$ with $m_e = 0$) and performing the polarization sums over the massive W bosons ($\sum \varepsilon^\mu \varepsilon^{*\nu} = -g^{\mu\nu} + k^\mu k^\nu / m_W^2$), one obtains:

$$\begin{aligned} \langle |\mathcal{M}|^2 \rangle &= \langle |\mathcal{M}_t|^2 \rangle + \langle |\mathcal{M}_u|^2 \rangle + 2 \langle \text{Re}(\mathcal{M}_t^* \mathcal{M}_u) \rangle \\ &= \frac{16G_F^2 m_\nu^2 (s - 2m_W^2)}{t^2 (s + t - 2m_W^2)^2} \\ &\quad \times \left[2m_W^8 - m_W^6 (5s + 8t) + 2m_W^4 (s^2 + 5st + 6t^2) - m_W^2 t (3s^2 + 11st + 8t^2) + 2t^2 (s + t)^2 \right] \end{aligned}$$

where we used $g_W^4/64 = G_F^2 m_W^4/2$.

Total Cross Section:

The t -variable in the CM frame is:

$$t = (p_1 - k_1)^2 \simeq m_W^2 - \frac{s}{2} \left(1 - \sqrt{1 - \frac{4m_W^2}{s}} \cos \theta \right)$$

The differential cross section:

$$\frac{d\sigma}{d\Omega} = \frac{\langle |\mathcal{M}|^2 \rangle}{64\pi^2 s} \frac{|\vec{k}_1|}{|\vec{p}_1|} = \frac{\langle |\mathcal{M}|^2 \rangle}{64\pi^2 E_{cm}^2} \sqrt{1 - \frac{4m_W^2}{E_{cm}^2}}$$

Integrating over the polar angle, the total cross section is:

$$\begin{aligned} \sigma &= \frac{6G_F^2 m_\nu^2}{\pi E_{cm}^4 (E_{cm}^2 - 2m_W^2)} \times \\ &\quad \left[E_{cm} \sqrt{E_{cm}^2 - 4m_W^2} (E_{cm}^4 - 3m_W^2 E_{cm}^2 + 2m_W^4) + 2m_W^2 (E_{cm}^4 - 2m_W^2 E_{cm}^2 + 2m_W^4) \tanh^{-1} \left(\frac{E_{cm} \sqrt{E_{cm}^2 - 4m_W^2}}{E_{cm}^2 - 2m_W^2} \right) \right] \end{aligned}$$

Numerical Estimate:

Using:

$$m_W = 80.377 \text{ GeV} \quad m_\nu = 0.8 \text{ eV} \quad \sqrt{s} = E_{cm} = 500 \text{ GeV}$$

where $m_\nu \lesssim 0.8 \text{ eV}$ is the current KATRIN upper bound, and $\sqrt{s} = 500 \text{ GeV}$ is representative of a future e^-e^- collider (ILC-scale):

$$\sigma \sim \mathcal{O}(10^{-56}) \text{ cm}^2 \sim 10^{-17} \text{ fb}$$

Even with the most optimistic projected luminosity of $\mathcal{L} \sim 10^4 \text{ fb}^{-1}$:

$$N_{\text{events}} = \mathcal{L} \times \sigma \sim 10^4 \times 10^{-17} = 10^{-13}$$

Thus, it is **not a realistic prospect** to observe $e^-e^- \rightarrow W^-W^-$.

Why It Is Hopeless — Scaling Argument:

Since $\sigma \propto m_\nu^2$, producing even a single event ($N \geq 1$) with $\mathcal{L} \sim 10^4 \text{ fb}^{-1}$ requires $\sigma \gtrsim 10^{-4} \text{ fb}$, i.e. a neutrino mass of:

$$m_\nu \gtrsim m_\nu^{\text{actual}} \times \sqrt{\frac{10^{-4} \text{ fb}}{\sigma_{\text{actual}}}} \sim 0.8 \text{ eV} \times \sqrt{\frac{10^{-4}}{10^{-16}}} \sim 0.8 \times 10^6 \text{ eV} \sim \mathcal{O}(\text{MeV})$$

This is six orders of magnitude above the actual neutrino mass scale. The extreme suppression has a transparent physical origin: the amplitude $\mathcal{M} \propto m_\nu$ because the Majorana mass must simultaneously (1) provide the $\Delta L = 2$ lepton number violation and (2) flip the chirality between the L - and R -projectors at the two vertices.

The most promising experimental probe of Majorana neutrino masses remains neutrinoless double beta decay ($0\nu\beta\beta$), where nuclear matrix elements provide enhancement factors absent in the elementary $e^-e^- \rightarrow W^-W^-$ process.

(see `task-4-4.m`)